University Database Relationship Documentation

The university database system is designed to reflect the hierarchical and functional structure of a typical institution of higher learning, such as Meru University. At the highest level, the University entity represents the institution itself and serves as the parent to various Schools (or Faculties). Each University can have multiple Schools, such as the School of Computing, School of Engineering, or School of Business. This relationship is implemented as a one-to-many association, where each School is linked to a single University via a foreign key.

Each School is further divided into several Departments. For example, the School of Computing may include the Department of Computer Science and the Department of Information Technology. This is also a one-to-many relationship: each School oversees multiple Departments, and each Department belongs to only one School.

Within a Department, there are multiple Lecturers, Courses, and Units (also called subjects or modules). A one-to-many relationship links the Department to Lecturers - each Department employs multiple Lecturers, but each Lecturer is associated with one Department. Similarly, each Department offers various Courses (such as BSc in Computer Science, Diploma in IT, etc.), with each Course tied to one Department.

Courses represent academic programs that students enroll in, typically lasting several years. Each Student belongs to one Course, forming another one-to-many relationship, where a Course can have many Students, but a Student is enrolled in only one Course.

The Unit entity represents specific subjects offered within a Department - for instance, "Database Systems" or "Operating Systems." Each Unit is offered by a single Department and is taught by a single Lecturer, establishing a many-to-one relationship between Unit and both Department and Lecturer.

The enrollment of students into specific units is managed through the Enrollment table, which acts as a junction (link) table. This allows for a many-to-many relationship between Students and Units, since a single Student can register for multiple Units, and a Unit can be taken by multiple Students. The Enrollment table includes additional attributes such as semester, academic year, and grade, which capture contextual and performance data about the student's participation in the unit.

This design ensures data normalization, supports scalability, and maintains logical consistency across all academic operations in the university - from structuring academic programs to tracking student progress, managing lecturers, and organizing teaching units across departments and schools.